Local model accuracy
Before & after fixes

4 November 2026 | Research_Paper.pdf | 20 fixed development questions

Exploratory, unblinded development-set regression test. These counts describe this document and question set, not general model accuracy.

Strictly correct answers

Model / output	Before / 20	After / 20
MiniCPM5 1B Q8_0	11/20 (55%)	9/20 (45%)
Gemma 4 E4B Q4_K_M (7.5B total)	8/20 (40%)	10/20 (50%)
Combined output	6/20 (30%)	13/20 (65%)

The final run returned all 20 combined responses. Two Gemma selections exhausted their output-token limit; the combined outputs retained valid MiniCPM excerpts. The baseline completed 19 requests; one connection failure is counted unsuccessful in the table. Completed-response baseline rates: 11/19, 8/19, 6/19.

Interpretation

The output policy changed: the repaired system selects original source excerpts rather than freely rewriting model answers. This prevents invented rewrites, but irrelevant selections, missing facts and unnecessary refusals still fail the rubric. Verification means cited text matched, not that the question was answered correctly.

Changes and test scope

Retrieval and extraction

Preserved PDF table layout, fixed overlap boundaries, and added lexical candidates plus vector/BM25 rank fusion. Existing stored documents require reingestion. Workspace and deletion predicates remain in both SQL candidate channels.

Grounding and synthesis

Each model selects evidence span IDs using a constrained JSON response. Code renders source text and attaches citations. The default combiner revalidates and deduplicates the selected source spans without a third model call. Invalid responses are rejected; failed candidates do not supply facts.

Bounded verification

Exact source excerpts are matched directly to their cited passages. Other claims have a 64-token / 15-second judge limit. Ambiguous, repetitive, failed or incorrectly cited verdicts cannot pass. The initial 2,000-token repetitive-verifier attempt is preserved separately.

What actually ran

Two downloaded LM Studio checkpoints, the local Nomic embedder, and application extraction, chunking, evidence selection, generators, synthesis and corrected verification. The retrieval index was isolated in memory. Go auth, PostgreSQL hybrid SQL execution, MinIO and full deployment were not tested.

Comparison limits

The same questions informed repair work: this is a development regression set, not held-out validation. Multiple settings changed together, including 0.3 to 0.0 temperature, three to four evidence chunks, and free text to extractive answers. One trial per question, with unblinded assistant scoring. No claim of causal attribution or general superiority is justified.

Runtime

Baseline median/p95 question wall time: 20.8s / 38.1s. After: 6.2s / 8.8s. These include shared fan-out and synthesis; model loading and indexing are excluded. The baseline omitted verification; the corrected run includes it.

Question-level results / 1

ID / question	Before M / G / C	After M / G / C
Q01 What decisions does the manuscript's evidence gate make at scores 0.29, 0.30, and 0.45?	Yes / No / No	Yes / Yes / Yes
Q02 Does the manuscript's deterministic evidence threshold estimate the probability that an answer is correct?	Yes / No / No	Yes / Yes / Yes
Q03 What are the manuscript's default chunk-window size and nominal chunk overlap, in characters?	No / No / No	Yes / No / Yes
Q04 What vector dimensionality does the manuscript use for stored embeddings?	Yes / Yes / No	Yes / Yes / Yes
Q05 According to the manuscript, is Redis a demonstrated durable job transport, and is a durable retry worker implemented?	Yes / Yes / No	Yes / Yes / Yes
Q06 Which workspace role can approve a report, and what task state pauses for that approval?	Yes / No / No	No / No / Yes
Q07 What temperature and maximum output-token count does the manuscript's normal task-generation path use?	No / No / No	Yes / Yes / Yes
Q08 What does verified=true mean for answer memory in the manuscript, and how much generated text can be stored?	Yes / Yes / Yes	Yes / No / Yes
Q09 How many authenticated-read requests were tested in total, how many per stage, and how many errors were observed?	No / No / No	No / No / No
Q10 What was the peak authenticated-read throughput, and at how many concurrent clients did it occur?	No / No / No	No / No / No

M = MiniCPM5 1B; G = Gemma E4B (7.5B total); C = combined source selection. Yes means all requested facts are correct with no material false additions. Citation formatting alone is not a pass.

Question-level results / 2

ID / question	Before M / G / C	After M / G / C
Q11 What p95 latency did the 12-task AI workload record, and does the API-read throughput establish thousands of generated answers per second?	No / No / No	No / Yes / Yes
Q12 Does the earlier three-question pilot's mean keyword coverage of 1.0 establish general answer accuracy?	Yes / Yes / Yes	Yes / Yes / Yes
Q13 Which two model tags and two temperatures were used in the manuscript's fixed-evidence pilot, and how many measured generations were run?	No / No / No	No / Yes / Yes
Q14 Which model-temperature-seed trial adopted the unsupported deletion premise, and did that trial actually delete documents?	No / No / No	No / No / No
Q15 Is this manuscript a peer-reviewed accepted paper, and have named authors and affiliations been finalized?	Yes / Yes / Yes	No / No / No
Q16 Does the manuscript establish an air-gapped or legally certified environment just because the workbench is called sovereign?	Yes / Yes / Yes	Yes / No / Yes
Q17 What is the exact annual commercial subscription price of this workbench in Indian rupees?	No / Yes / No	No / Yes / No
Q18 What is the patent application number filed by this project's inventors?	Yes / Yes / Yes	No / No / No
Q19 How many real industrial customers have deployed this workbench in production?	Yes / No / Yes	No / No / No
Q20 The paper proves a general autonomous DAG planner. Which general DAG planning algorithm was validated?	No / No / No	No / Yes / Yes

M = MiniCPM5 1B; G = Gemma E4B (7.5B total); C = combined source selection. Yes means all requested facts are correct with no material false additions. Citation formatting alone is not a pass.

Review and reproducibility

What remains unresolved

Q09: Retains the available facts but does not explicitly answer the requested total; the user would have to calculate it.

Q10: Retains the client count but not the required throughput value.

Q14: Retains the trial identity but still does not explicitly answer the actual-deletion question.

Q15: Retains pending author details but does not explicitly answer publication/peer-review status.

Q17: Retains the irrelevant provenance excerpt and does not explicitly abstain; no price is invented.

Q18: Does not answer or explicitly abstain; it does not invent a patent number.

Q19: Retains irrelevant excerpts rather than explicitly abstaining about customers.

Files and model identities

Baseline folder: tests/evaluation/results/lmstudio-document-2026-11-03
Final folder: tests/evaluation/results/lmstudio-document-2026-11-04-final
Full answers and review notes: transcript.md and review.json in the final folder. Exact prompts and responses: http_calls.jsonl. Source/document/question hashes: manifest.json.

MiniCPM5-1B Q8_0 (openbmb) and gemma-4-E4B-it-OBLITERATED Q4_K_M (OBLITERATUS). LM Studio reports the latter as 7.5B total parameters. The existing Nomic Embed Text v1.5 Q4_K_M embedder was used. These exact downloaded checkpoints were tested, not every model in their families.

A second document and independently labeled held-out questions are needed before a broader accuracy claim. Source accuracy, recall, completeness and answer relevance still require evaluation. The database SQL path and a full authenticated deployment remain separate checks.